

Week 3 Corrected Report

Behaviour Dataset Construction and Visualization

Corrected scan-window version using manually concatenated observation intervals and segment-to-Excel-window alignment.

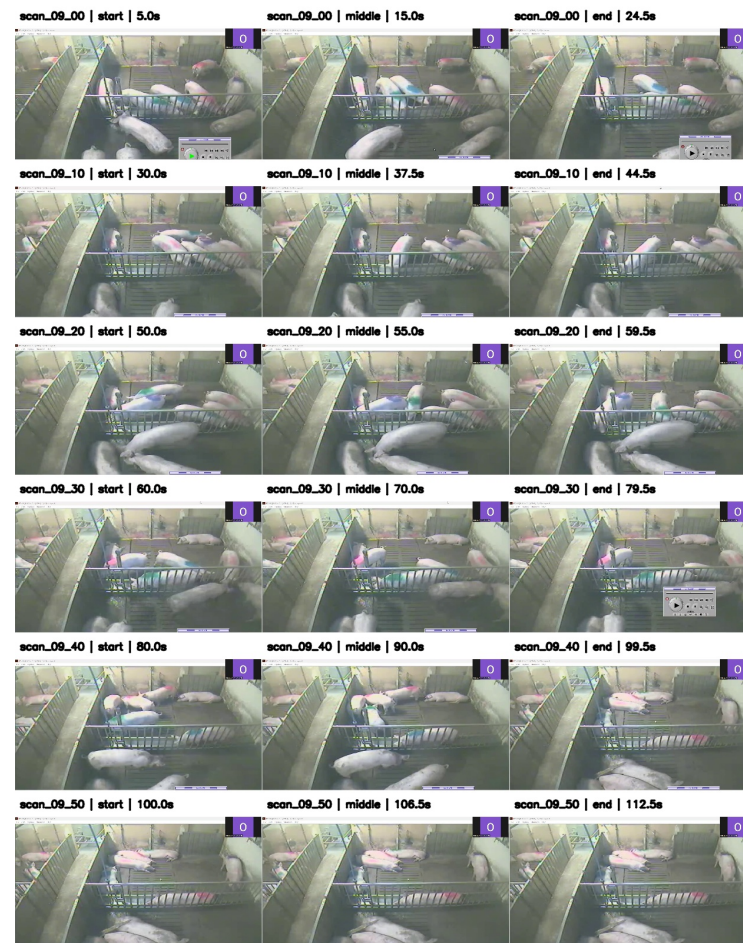
Main correction: the previous no-overlap limitation between video and Excel scan-sampling windows was addressed.

Remaining conservative choice: track-to-colour identity mapping is kept as a manual verification step to avoid false behaviour labels.

1. Purpose of the correction

The first Week 3 version successfully built the behaviour annotation structure, but the original video sample did not exactly overlap with the Excel scan-sampling observation windows. To fix this, a new clean screen-recording video was prepared by placing the relevant observation windows one after another.

The corrected version uses six aligned scan windows: 09:00, 09:10, 09:20, 09:30, 09:40, and 09:50. Each segment is manually mapped to its corresponding Excel interval.



2. Corrected processing pipeline

Step	Description
1	Check clean scan-window video metadata and sample frames.
2	Create manual segment-to-Excel-window mapping.
3	Extract each scan-window segment into a separate image sequence.
4	Run YOLOv8-s detection on each segment.
5	Run ByteTrack tracking on each segment.
6	Attach segment ID and Excel interval information to each track row.
7	Parse Excel behaviour labels for each scan window.
8	Build corrected JSON annotation dataset.
9	Generate visualization videos and package final outputs.

3. Corrected JSON dataset summary

The corrected JSON links video frame, track ID, bounding box, segment ID, Excel interval, and segment-level behaviour candidates.

segment_id	excel_interval_start	excel_interval_end	frames	track_instances	verified_track_instances	unverified_track_instances
scan_09_00	09:00:00	09:00:10	500	5139	0	5139
scan_09_10	09:10:00	09:10:10	375	3294	0	3294
scan_09_20	09:20:00	09:20:10	250	2682	0	2682
scan_09_30	09:30:00	09:30:10	500	4968	0	4968
scan_09_40	09:40:00	09:40:10	500	4481	0	4481
scan_09_50	09:50:00	09:50:10	325	2670	0	2670

4. Tracking summary

ByteTrack was applied separately to each extracted scan-window sequence. Some segments have more unique track IDs than the number of pigs because of ID fragmentation, partial visibility, neighbouring pen detections, and screen-recording artefacts.

segment_id	frames	track_rows	unique_track_ids	excel_interval_start	excel_interval_end
scan_09_00	500	5139	33	09:00:00	09:00:10
scan_09_10	375	3294	19	09:10:00	09:10:10
scan_09_20	250	2682	19	09:20:00	09:20:10
scan_09_30	500	4968	32	09:30:00	09:30:10
scan_09_40	500	4481	15	09:40:00	09:40:10
scan_09_50	325	2670	10	09:50:00	09:50:10

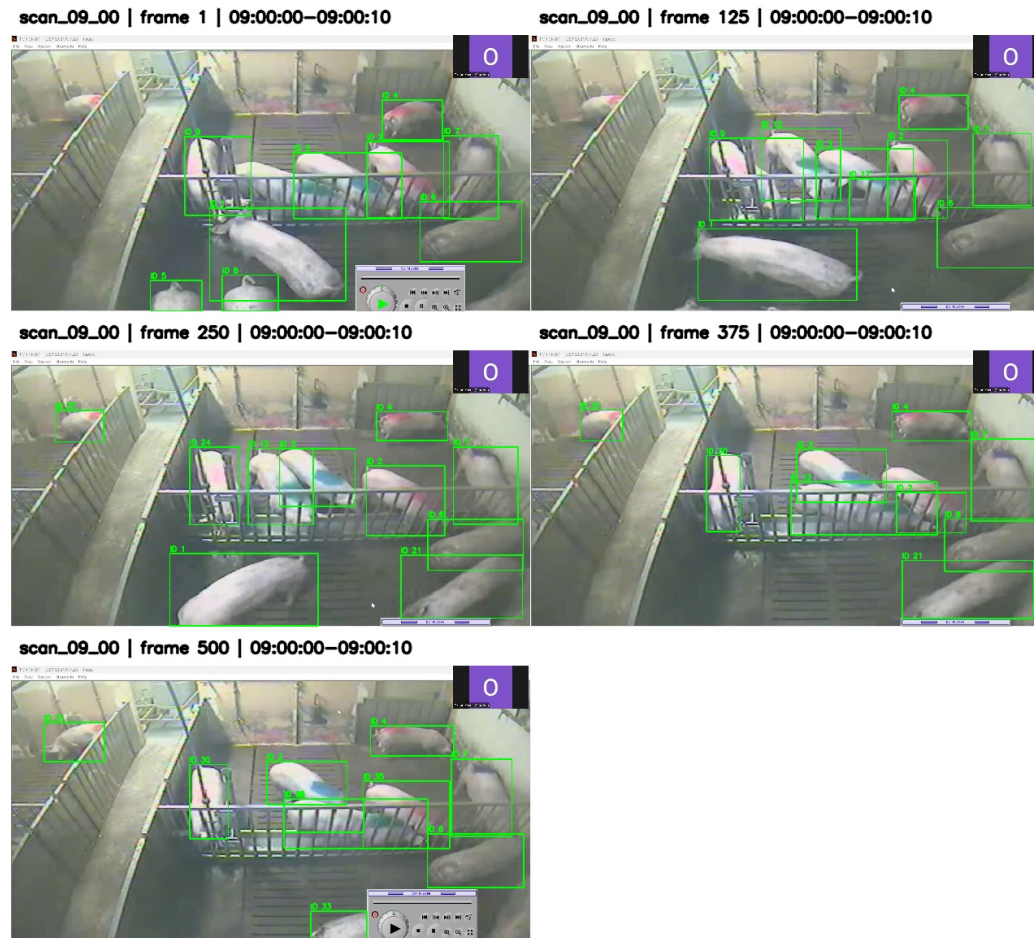
5. Behaviour labels from Excel

The Excel file provides scan-sampling behaviour labels for colour-coded pig identities. These labels are now correctly attached to the corresponding scan-window segments.

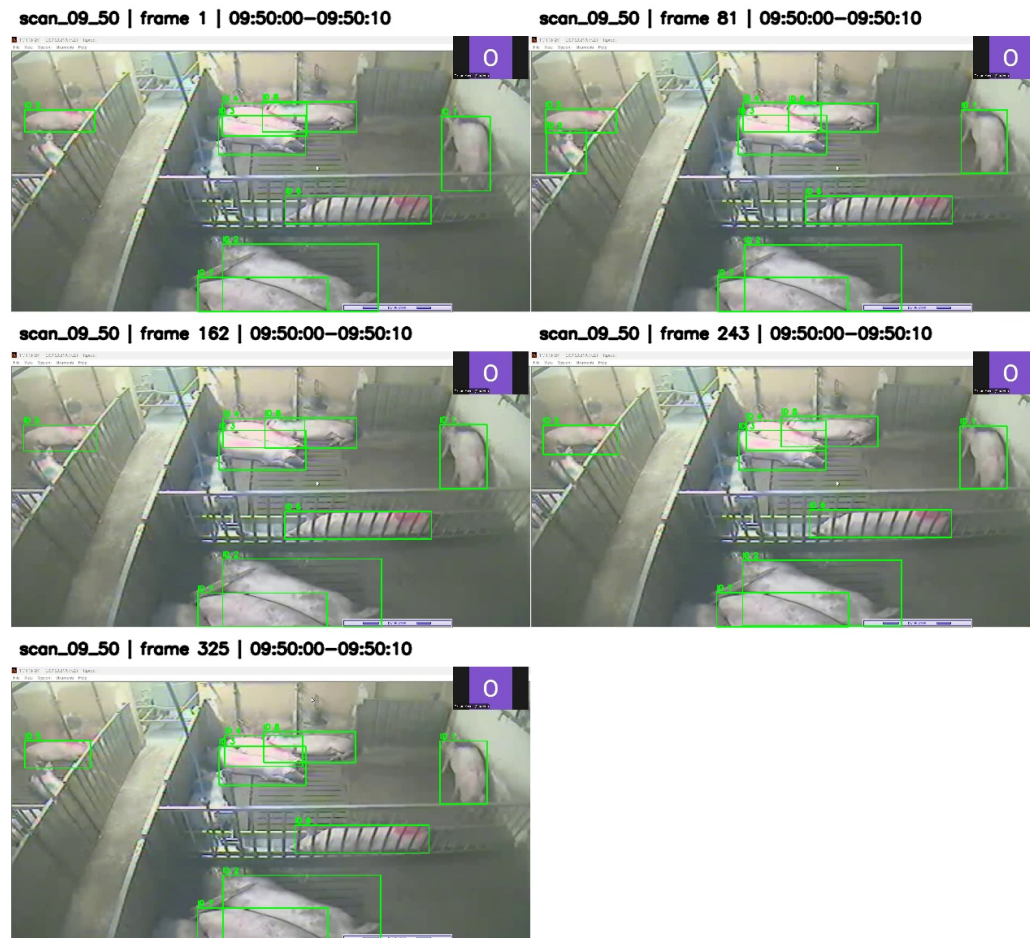
segment_id	excel_interval_start	excel_interval_end	num_pigs	colours	behaviour_codes	behaviour_labels
scan_09_00	09:00:00	09:00:10	6	green, blue, purple, red neck, red tail, no color	NU, NU, NU, NU, AN, NU	eating, eating, eating, eating, exploring_or_sniffing, eating
scan_09_10	09:10:00	09:10:10	6	green, blue, purple, red neck, red tail, no color	BOX, BOX, BOX, BOX, STI, BOX	out_of_view_or_box, out_of_view_or_box, out_of_view_or_box, out_of_view_or_box, lying_sternal, out_of_view_or_box
scan_09_20	09:20:00	09:20:10	6	green, blue, purple, red neck, red tail, no color	BOX, BOX, PI, BOX, STI, AN	out_of_view_or_box, out_of_view_or_box, standing_inactive, out_of_view_or_box, lying_sternal, exploring_or_sniffing
scan_09_30	09:30:00	09:30:10	6	green, blue, purple, red neck, red tail, no color	BOX, BOX, STI, BOX, STI, BOX	out_of_view_or_box, out_of_view_or_box, lying_sternal, out_of_view_or_box, lying_sternal, out_of_view_or_box
scan_09_40	09:40:00	09:40:10	6	green, blue, purple, red neck, red tail, no color	AN, BOX, STI, AN, IN, AN	exploring_or_sniffing, out_of_view_or_box, lying_sternal, exploring_or_sniffing, interaction, exploring_or_sniffing
scan_09_50	09:50:00	09:50:10	6	green, blue, purple, red neck, red tail, no color	LAI, STI, PI, STI, STI, STI	lying_lateral, lying_sternal, standing_inactive, lying_sternal, lying_sternal, lying_sternal

6. Track ID montage examples

Track ID montages were generated for every scan window. These figures support manual verification of track-to-colour identity mapping.



7. Additional montage example



8. Identity mapping policy

The tracker produces numerical IDs such as track_id=1 or track_id=2. However, the Excel file gives labels for colour-coded identities such as green, blue, purple, red neck, red tail, and no color. Therefore, track-to-colour mapping is required before assigning a final behaviour label to an individual track.

This corrected version keeps individual track identities unverified unless a human manually confirms the colour identity. This conservative approach avoids assigning incorrect behaviour labels when colour markers are unclear or when ID fragmentation occurs.

segment_id	track_id	first_frame	last_frame	num_frames	mean_score	mean_cx	mean_cy
scan_09_00	6	1	500	500	0.9038	1706.8	724.5
scan_09_00	4	1	500	500	0.899	1481.05	284.45
scan_09_00	7	1	500	500	0.8862	1730.88	493.7
scan_09_00	3	1	500	495	0.8342	1165.34	493.26
scan_09_00	12	8	500	492	0.8126	1107.99	545.5
scan_09_00	2	1	478	478	0.822	1458.84	558.48
scan_09_00	1	1	341	341	0.9238	936.2	873.95
scan_09_00	21	163	388	226	0.8537	1646.11	879.21
scan_09_00	30	319	500	182	0.6155	721.79	512.07
scan_09_00	9	1	174	174	0.7141	793.48	514.06
scan_09_00	31	321	500	174	0.6433	241.41	280.14
scan_09_00	24	188	335	142	0.5525	761.48	495.63
scan_09_10	7	1	375	375	0.8666	1486.6	591.87
scan_09_10	2	1	375	375	0.8091	1154.31	952.84
scan_09_10	4	1	375	375	0.7927	1585.34	381.41
scan_09_10	3	1	375	375	0.6859	1810.43	742.66
scan_09_10	5	1	375	375	0.5648	693.89	881.36
scan_09_10	14	88	375	288	0.5659	1541.98	452.06

9. Final outputs and conclusion

The corrected final package contains corrected JSON annotations, segment-level behaviour labels, tracking CSV files with Excel window information, identity mapping templates, montages, visualization videos, scripts, and summary notes.

The main previous limitation, no exact overlap between video and Excel scan windows, has been fixed. The remaining limitation is identity verification: individual track IDs still require manual confirmation before they can be safely linked to colour-coded pig identities.

Overall, the corrected Week 3 dataset now provides a stronger behaviour annotation foundation by linking frames, tracked pig instances, scan-window timing, and Excel behaviour labels in a structured JSON format.